

NILANJAN CHATTERJEE

Principal Data Architect | Enterprise Lakehouse, Data Mesh & Fabric | Governance, Contracts & MLOps
1-704-345-9929 | nilanjan.9325@gmail.com | linkedin.com/in/nil68657 | github.com/nil68657

PROFESSIONAL SUMMARY

Principal Data Architect with 13+ years architecting petabyte-scale enterprise data platforms across silicon, financial services, retail, telecom, and cloud-services domains. Deep expertise in target-state Lakehouse, Data Mesh, and Data Fabric design; data contracts, data products, and federated governance; real-time and batch integration patterns; and operationalizing MLOps / LLMOps at scale. AWS, GCP, Databricks, and Snowflake certified, with hands-on fluency in Anthropic Claude, OpenAI GPT-4o, LangChain, and LangGraph. Proven driver of 0-to-1 greenfield platforms and 1-to-100 scale-out programs, delivering \$5M+ in measurable revenue / cost outcomes, p9999 SLA compliance, and 85% YoY Sev-2 reductions across multi-cloud landscapes. Trusted advisor to CDO, CTO, and product leadership on enterprise data strategy and architecture review boards.

PROFESSIONAL EXPERIENCE

Advanced Micro Devices (AMD) • Principal Staff Data Architect • Austin, TX • July 2024 – Present

- Owns AMD's **enterprise data target-state architecture** — a Unified Lakehouse with federated Data Mesh and Data Fabric overlays — integrating dozens of internal silicon-validation, manufacturing, automation, telemetry, and external SaaS sources for the **Radeon, Instinct, and Helios** product portfolios.
- Architected production data-serving infrastructure on **CDP using Apache Kafka, OneHouse, Aerospike, Trino, Dremio, Apache Spark, and Elasticsearch**, sustaining **sub-millisecond p9999 latency SLAs** across 2 PB/quarter of telemetry and feature data.
- Designed the enterprise **Lakehouse and governance** plane on **Databricks (Delta Lake, DLT, Unity Catalog, LakeBase, AgentBricks)** and **Snowflake (Dynamic Tables, Cortex, OpenFlow)** supporting **Lambda | Kappa | Delta** Data Mart patterns for pre/post-silicon validation, utilization, and yield analysis DataOps and MLOps pipelines.
- Defines and stewards **data contracts, semantic models, and data-product specifications** with leading SaaS and Cloud partners; runs the org's Architecture Review Board (ARB) and 2-week design-review cadence.
- Engineered LLM-assisted **agentic DataOps** using **Anthropic Claude (Opus / Sonnet), OpenAI GPT-4o, LangChain, and LangGraph** to automate data-quality checks, pipeline monitoring, and anomaly RCA — reducing manual triage by **60%**.
- Built **ML monitoring and telemetry framework** for data, performance, and inference drift using **MLflow, BentoML, Prometheus, Dagster, Grafana, Loki, Mimir, and Thanos**; powered the BI surface via **Databricks SQL and Apache Superset**, delivering **\$3M+ YoY** revenue growth.
- Established event-driven, multi-tier lakehouse patterns with HA/DR, reducing **Sev-2 incidents by 85% YoY** with **GitHub Copilot and Databricks Agents**; operationalized governance and compliance — **PCI DSS, HIPAA, GDPR, CCPA, GLBA, NIST 800-53, SOC 2, ISO 27001** — via **LightUp, Alation, Monte Carlo, and Collibra**.
- Architected **HPC multi-node GPU clusters** supporting **GenAI, RAG, and RLHF** pipelines with **vLLM and TGI** for efficient SOTA model serving on the ROCm + LLVM/MLIR stack.
- Leads 12+ SDEs, Data Engineers, and AI/ML Scientists on Power Management, Performance Logging, Manufacturing, and Test Automation programs.

TransUnion • Principal Data Architect • Austin, TX • Apr 2023 – July 2024

- Set the enterprise data architecture and Go-to-Market strategy across fraud, device, credit, non-credit risk, cyber, and churn product lines; led **18 Data Engineers, Data Scientists, and SDEs** delivering 0-to-1 turn-key platforms and 1-to-100 scaled rollouts.
- Architected the **Customer Data Platform (CDP)** and **Data Cleanroom** target state supporting **Lambda | Kappa** Data Marts and the full **MLOps lifecycle** (registry, serving, monitoring, governance); integrated GenAI / RAG and lifted fraud-identification accuracy by **32% YoY** via identity keying-matching-linking pipelines on **Feast, Redis, Aerospike, dbt, MLflow, BigQuery, Airflow, Pub/Sub, Kubernetes, and Trino**.
- Designed the **Aerospike-backed online/offline Feature Store** on **Feast, Tecton, and Kubeflow** — cut Scoring/Fraud/AML inference latency from **35 ms → 7 ms (80%)** and saved **\$1.2M/year** in serving cost.
- Architected the **multi-cloud MLOps serving plane** (GCP, AWS, Azure) on **Cloud Spanner, Terraform, Kong, and Swagger** with GraphQL/REST endpoints over **Parquet, Protobuf, Apache Arrow, and Apache Iceberg**, generating **\$1M+ YoY** in net-new product revenue.

- Designed **A/B and A/A testing** patterns with custom traffic bifurcation, CUPED variance reduction, and SPRT early-stopping for credit and non-credit feature rollouts.
- Architected privacy-preserving **Data Cleanrooms** with anonymized / PET-enabled assets integrated with knowledge and identity graphs; defined cross-tenant access policies.
- Operationalized cost-efficient lakehouse and orchestration patterns using **Kafka, Flink, Spark, Hudi, Iceberg, Airflow, S3, Redshift, TimescaleDB, and Aerospike** with Avro/Parquet serialization.
- Architected distributed **GenAI/RAG** services on **GPT-4 and Llama** with **Milvus, LangChain, and LlamaIndex**, retiring legacy NLP infrastructure and saving 10+ enterprise clients **\$2.5M+** annually.
- Engineered low-latency threat / intrusion-detection ingestion pipelines on **CrowdStrike, SIEM, Cloudflare, Splunk, Filebeat, and ELK**.

Amazon • **Sr. Data Architect** • *Arlington, VA* • *July 2020 – Apr 2023*

- Architected enterprise **data strategy and roadmap** with **Lambda | Kappa | Delta** patterns over **Lakehouse | Lake | Mesh | Mart** designs across **Redshift, Aurora, Kinesis, Firehose, DynamoDB, dbt, Fivetran, Athena, and Airflow** for FinTech, Education, Retail, and FMCG clients — reducing data-blocker delivery misses by **90%**.
- Led a team of 6 BIEs, Data Scientists, and Data Engineers across multiple analytics customer engagements; ran weekly reviews and growth-tracking that drove direct-report promotions.
- Improved query latency by **70%** via **Apache Iceberg with Project Nessie** and optimized data-lake caching, saving **400 resource-hours / year**.
- Established enterprise **data quality, SLA, and resolution frameworks** ensuring data accuracy from creation through presentation layers — reducing **Sev-2 incidents by 50% annually**.
- Architected **ML/AI/BI roadmaps** on **Spark, EMR, Glue, Redshift, Elasticsearch, and Kibana** to serve analytics and data science needs for client engineering teams.
- Architected the Amazon Seller / Buyer **capacity-planning attribution model** end-to-end (onboarding → shipment), boosting operational performance by **200 bps** with proactive RBAC-based delivery (Kinesis, Glue, DynamoDB).
- Built Airflow with multi-executor DAGs for real-time processing integrating **Kafka, Hudi, and Flink**; migrated legacy cron + Jenkins jobs and improved failover with idempotent operators.
- Formulated statistical runbooks for pre/post A/B and A/A test analysis and ML pipeline productionization (SageMaker, Lambda, DynamoDB); implemented Redshift metadata management and automated Glue Crawler pipelines.

Hughes Network Systems • **Data Mining Scientist (Tech Lead)** • *Germantown, MD* • *Jan 2018 – June 2020*

- Led 5 Data Engineers and Analysts building edge-served ML models for the two largest US telecom providers; reduced national audio call-drop rate by **72%** and delivered **\$650K+ QoQ** revenue growth.
- Built ML-driven CDN caching and frame-rate prediction models, cutting frame drops by **48%** and MoM churn by **26%** on low-bandwidth networks.
- Engineered NLP/NLU pipelines on **AWS SageMaker, Textract, spaCy, gensim, AllenNLP, ELMo, and RoBERTa** for multi-LOB survey text summarization; built RL/ARIMA pipelines on **PySpark, Kafka, Glue, Lambda, Hive, and NiFi**.
- Designed warehouses and lakes on **Oracle 11g/12c (Data Guard, ASM, RMAN, RAC)**; reduced supply-chain assortment bottlenecks by **4%** lifting profit by **6%**.

IBM Corporation • **Data Science Intern** • *Durham, NC* • *June 2017 – Dec 2017*

- Built BlueMix usage-pattern feature pipelines and ML models — **PCA, SVD, Market Basket Analysis, kNN, GBM, Hierarchical Clustering, Logistic Regression with Bayesian Belief, MCMC** — and rolled them out via challenger-vs-champion A/B testing with the Sales/Pre-Sales team.
- Designed model deployment and optimization strategy using **CPLEX, Gurobi, and LAPack** on **Spark MLlib**.

Cognizant Technology Solutions • **Senior Data Engineer** • *Bangalore, India* • *July 2013 – Aug 2016*

- Tech-led 3 Data Engineers building **Informatica, Talend, Cognos BI, and SAP Business Objects** ETL pipelines for the Marketing org, delivering **\$500K YoY** revenue growth.
- Engineered a **50-node Apache Spark** stream-processing cluster for **10TB+/day** with **Kafka, Flume, Impala, HBase, and Oozie**; migrated DWs and data lakes to a 50-node Hadoop cluster (Pig, Hive, HBase, Impala, Ambari, Mahout).
- Delivered AWS / Azure cloud integrations using **S3, EMR, DynamoDB, Redshift, and Lambda** following CRISP-DM; implemented CI/CD with **Git, Maven, Jenkins, Chef, Ansible, and Puppet**.

- Managed DW integration with Erwin Data Modeler, MSSQL, SSIS, SSAS, SSRS, PL/SQL, and T-SQL; presented analytics outcomes in MBR and QBR forums.

EDUCATION

M.S., Computer Science — University of North Carolina at Charlotte, NC, USA | 2016 – 2017 | GPA 3.8 / 4.0

B.S., Electronics & Communication Engineering — West Bengal University of Technology, India | 2009 – 2013 | GPA 3.7 / 4.0

CERTIFICATIONS

Databricks Certified Data Engineer Professional | AWS Certified Machine Learning – Specialty | AWS Certified Developer – Associate | SnowPro Core Certified | GCP Professional Data Engineer

TECHNICAL SKILLS

Languages: Python, SQL, Scala, Java, Rust, C++, PySpark, Bash

Cloud Platforms: AWS (S3, Glue, Redshift, EMR, Lambda, SageMaker, Bedrock, Kinesis, Athena, DynamoDB, EKS), GCP (BigQuery, Dataflow, Pub/Sub, Vertex AI), Azure (ADF, Synapse, ADLS, Event Hubs), Oracle Cloud

Data Lakehouse: Databricks (Delta Lake, DLT, Unity Catalog, LakeBase, AgentBricks), Snowflake (Dynamic Tables, Cortex, OpenFlow), Apache Iceberg, Apache Hudi, OneHouse, Delta Sharing, Apache Polaris/Nessie

Architecture Patterns: Data Mesh, Data Fabric, Data Lakehouse, Medallion (Bronze/Silver/Gold), Lambda | Kappa | Delta, CQRS, Event-Sourcing, CDC, Outbox, Saga, Reverse-ETL, Data Products & Contracts

Streaming & Pipelines: Apache Kafka, Apache Flink, Apache Spark, Airflow, Dagster, Mage, NiFi, Pub/Sub, Kinesis Firehose, dbt, Fivetran, Airbyte, Debezium CDC

Databases & Stores: PostgreSQL, MySQL, Oracle, MongoDB, Cassandra, Redis, DynamoDB, Aerospike, BigQuery, Redshift, Trino, Dremio, DuckDB, TimescaleDB, Elasticsearch

AI / LLM Frameworks: Anthropic Claude API (Opus/Sonnet/Haiku), OpenAI GPT-4o / o1, Meta Llama 3, Mistral, DBRX, LangChain, LangGraph, LlamaIndex, Hugging Face, PEFT/LoRA, CrewAI, AutoGen, Ollama, vLLM, TGI

Vector Databases: Pinecone, Milvus, ChromaDB, Weaviate, Qdrant, pgvector, FAISS

MLOps & Feature Stores: MLflow, BentoML, Feast, Tecton, Kubeflow, Prometheus, Grafana, Loki, Monte Carlo, OpenLineage, Great Expectations, Soda

DevOps & IaC: Docker, Kubernetes, Terraform, Helm, GitHub Actions, Jenkins, GitLab CI/CD, ArgoCD, Ansible

AI Coding Tools: Cursor, GitHub Copilot, Databricks Agents, Amazon CodeWhisperer

Governance & Compliance: Atlan, Alation, LightUp, Collibra, PCI DSS, HIPAA, GDPR, CCPA, NIST 800-53, SOC 2, ISO 27001, FedRAMP

NOTABLE PROJECTS

Healthcare Claims Intelligence Platform — Web-scraping and analytics pipeline collecting healthcare claims success-rate data across multiple providers; delivered an auto-refreshing Tableau executive dashboard. *Tech: Python, Scrapy, XGBoost, CatBoost, Tableau, Oracle, Airflow*

Quora Duplicate Question Detection — Character-level RNN + SVM ensemble using Word2Vec / Bag-of-Words representations, fine-tuned via GridSearchCV across TensorFlow and PyTorch. *Tech: Python, scikit-learn, Word2Vec, GloVe, TensorFlow, PyTorch*

Multi-Class Image Segregation Algorithm — Deep-learning classification pipeline using PyTorch and WEKA to automatically categorize and label unstructured images into fine-grained classes. *Tech: PyTorch, TensorFlow, NLTK, Apache Solr, Kafka, HBase*

Distributed PageRank on Apache Spark — Distributed PageRank with custom mapper/reducer allocations on Apache Spark for large-scale web graph analysis. *Tech: Apache Spark, Hadoop, Lucene, Solr, Java, Sqoop, Zookeeper*